MIL-P-60432C(AR)
1 DECEMBER 1980
SUPERSEDING
MIL-P-60432B(PA)
28 October 1974

MILITARY SPECIFICATION

PROPELLANT POWDER FOR USE IN BASE GRAIN, AHH

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication and packing of the propellant for use in the base grain, AHH.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-155	- Graphite (For Use in Ammunition)
MIL-C-206	- Cellulose, Cotton Linters
MIL-N-244	- Nitrocellulose (For Use in Explosives)
MIL-N-246	- Nitroglycerin
MIL-N-3399	- 2-Nitrodiphenylamine
MIL-L-13788	- Lead Salicylate
MIL-C-13609	- Casting Solvent for OGK, OIO and AHH
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-L-17699	- Lead 2-Ethyl Hexoate

☆U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/478

FSC 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-286 - Propellants, Solids Sampling, Examination and Testing

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposals, shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Department of Transportation: Parts 0-199

3. REQUIREMENTS

3.1 Materials. The material shall comply with the requirement of the applicable specification as follows:

MATERIAL	SPECIFICATIONS
Nitrocellulose	MIL-N-244, Grade A, Type I Made with Cellulose, Cotton Linters of MIL-C-206
Nitroglycerin	MIL-N-246, Type I
2-Nitrodiphenylamine	MIL-N-3399
Graphite	MIL-G-155, Grade III or IV
Lead Salicylate	MIL-L-13788
Lead 2-Ethylhexoate	MIL-L-17699

3.2 Composition. The composition shall conform to the requirements of Table I when tested as specified in 4.5.1.

TABLE I

MATERIAL	REQUIREMENTS PERCENT
Nitrocellulose	83.0 $\pm$ 1.3
Nitroglycerine	11.4 Nominal

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TABLE I (Cont'd)

MATERIAL	REQUIREMENTS PERCENT
Lead Salicylate	2.3 Nominal
Lead 2-Ethylhexoate	2.3 Nominal
2-Nitrodiphenylamine	1.0 ± 0.15
Graphite (Glaze)	0.03 to 0.06

3.3 Total Volatiles. The total volatiles shall not be greater than 1.50 percent when determined as specified in 4.5.3.

3.4 120° Heat Test. The powder shall not cause complete fading of the methyl violet paper to salmon pink color in less than 40 minutes and shall not explode in 5 hours when tested as specified in 4.5.4.

3.5 Residual Solvent. The residual solvent shall not be greater than 0.70 when determined as specified in 4.5.5.

3.6 Packing Density. The packing density of the powder shall be 0.95 min. when determined as specified in 4.5.6.

3.7 Grain Dimensions. Unless otherwise specified in the contract or purchase order, the grain dimensions when determined as specified in 4.5.7 shall be as follows:

Length: 0.035 ± 0.003 inches

Diameter: 0.035 ± 0.002 inches

The ends of the grains shall be substantially square and free of chips and tails.

3.8 Heat of Explosion and Strand Burning Rate on Cast Charges prepared in accordance with paragraph 4.5.2.1

3.8.1 Heat of Explosion. The heat of explosion shall be between 940 and 970 cal/gm when determined as specified in 4.5.2.3.

3.8.2 Strand Burning Rate

3.8.2.1 Burning Rate. The propellant shall meet the burning rate specified in Table II when determined as specified in 4.5.2.4.

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TABLE II

Pressure Pounds per Square Inch	Burning Rate - Inches per second	
	Lower Limit @ Minus 40°F	Upper Limit @ Plus 140°F
600	.359	.421
800	.414	.464
1000	.433	.450
1200	.368	.462
1400		.479
1600		.581
1800		.615
2000		.663

3.9 Workmanship. The best commercial practices shall be used in formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following type of inspection shall be conducted on this item:

a. Quality Conformance Inspection

4.3 Quality Conformance Inspection

4.3.1 Inspection Lot Formation. See MIL-A-48078.

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4.4 Examination and Tests. See MIL-A-48078.

4.4.1 Chemical Composition and Other Tests of Propellant. (3.2 through 3.7).

4.4.1.1 Sampling. Two one pound samples shall be selected at random from the lot of propellant. The samples shall be checked for the requirements specified in paragraph 3.2 through 3.7.

4.4.1.2 Rejection. If either sample fails to meet the requirements specified in paragraph 3.2 through 3.7, the lot of propellant shall be rejected.

4.4.2 Heat of Explosion on Cast Charges. The propellant powder shall be cast into three cast charges in accordance with paragraph 4.5.2.1. A sample shall be selected from each cast charge. The heat of explosion shall be determined on each sample selected from the three cast charges.

4.4.2.1 Rejection. If one of the samples selected from each of the cast charges fails to meet the requirements of 3.8.1, six additional cast charges shall be cast in accordance with paragraph 4.5.2.1. If any one of the additional cast charge samples fails to meet the requirement of 3.8.1, the lot of propellant powder shall be rejected.

4.4.3 Strand Burning Rate

4.4.3.1 Cast Charges. Thirty-six (36) strands shall be made in accordance with paragraph 4.5.2.4 from each of the three cast charges prepared per paragraph 4.5.2.1. Three strands shall be used to determine the average burning rate at each temperature pressure combination specified in paragraph 4.5.2.4.

4.4.3.2 Rejection. If any one of the average strand burning rates rails to meet the requirement of 3.8.2.1, six additional charges shall be cast in accordance with paragraph 4.5.2.1. Thirty-six (36) strands shall be made from each of these cast charges. Three strands shall be used to determine the average burning rate at each temperature pressure combination specified in paragraph 4.5.2.4. If any one of the average burning rates from each cast charge fails to meet the requirement of 3.8.2.1, the lot of propellant shall be rejected.

4.5 Test Methods and Procedures

4.5.1 Composition

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4.5.1.1 Nitrocellulose, Major Defect. The nitrocellulose content shall be determined in accordance with Standard MIL-STD-286, Method 209.6 or 209.2.

4.5.1.2 Graphite, Major Defect. The graphite content shall be determined in accordance with Standard MIL-STD-286, Method 308.1 or 309.1.

4.5.1.3 2-Nitrodiphenylamine, Major Defect. The 2-nitrodiphenylamine content shall be determined in accordance with MIL-STD-286, Method 218.4.

4.5.1.4 Nitroglycerin. The nitroglycerin content shall be determined in accordance with MIL-STD-286, Method 208.2 and the results recorded for information only.

4.5.1.5 Lead Salicylate. The lead salicylate shall be determined in accordance with MIL-STD-286, Method 225.1 and the results recorded for information only.

4.5.1.6 Lead 2-Ethyl Hexoate. The lead 2-ethyl hexoate content shall be determined in accordance with MIL-STD-286, Method 311.5 and the results recorded for information only.

4.5.2 Heat of Explosion and Strand Burning Rate from Cast Charge

4.5.2.1 Cast Charge

4.5.2.1.1 The cast charge shall be made to a minimum of three (3) inches in diameter. Actual size shall depend upon the availability of adequate casting molds. Both inhibited and non-inhibited cast charges are acceptable. However, in either case, the outer 1/4 inch peripheral layer of the cast charge shall not be utilized to obtain the required samples.

4.5.2.1.2 The AHH casting solvent composition shall comply with MIL-C-13609.

4.5.2.1.3 The AHH Base grain powder composition shall comply with 3.2.

4.5.2.2 Procedure for preparing the cast charge

4.5.2.2.1 The AHH casting solvent and the AHH base grain powder shall be evacuated at 10mm Hg maximum for 8 hours minimum.

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4.5.2.2.2 The powder shall be screened into the casting mold. A screening plate shall be placed 4 1/2 inches above the top of casting mold. This screening plate shall have 7/64 inch diameter holes on 5/32" centers. A distributor plate shall be placed 4 1/2 inches above the screening plate. Sufficient holes shall be drilled into the distributor plate so that the base grain powder will be evenly dispersed upon the screening plate when the base grain powder falls from the distributor plate on to the screening plate. The holes in the distributor plate shall be no less than .15 inch in diameter and the ratio between the end area of the cast charges to the total area of the distributor holes shall be 22/1. The powder shall be free to fall from the distributor plate, through the screening plate and into the casting mold without any obstructions.

4.5.2.2.3 The AHH casting solvent shall be flowed up through the packed casting mold. (Not applicable if using a top cast process).

4.5.2.2.4 The finished casting shall be cured for a suitable length of time to obtain a complete cure at 145°F.

4.5.2.3 Heat of Explosion, Major Defect. The heat of explosion shall be determined in accordance with Method 802.1 of MIL-STD-286.

4.5.2.4 Strand Burning Rate, Major Defect. The strand burning rate shall be determined in accordance with Method 803.1 of MIL-STD-286. Three (3) strands shall be tested from each cast charge under each pressure and temperature listed below:

Pressure (psi) at -40°F	Pressure (psi) at 140°F
600	600
800	800
1000	1000
1200	1200
	1400
	1600
	1800
	2000

4.5.3 Total Volatiles, Major Defect. The total volatiles shall be determined in accordance with Method 103.3 of MIL-STD-286.

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4.5.4 120°C Heat Test, Major Defect. The 120°C heat test shall be performed in accordance with Method 404.1 of MIL-STD-286.

4.5.5 Residual Solvent, Major Defect. The residual solvent content shall be determined in accordance with Method 103.4 or 103.5 of MIL-STD-286, except the moisture content shall be determined in accordance with Method 101.3 or with Method 102.1 when using Method 103.4.

4.5.6 Packing Density, Major Defect

4.5.6.1 Apparatus. The following pieces of apparatus shall be required: One 500 ml cylinder, three pieces of numbered sectional tubing, a hopper, distribution plate, screen loading plate, and a brass bar. All of these parts shall be made of normalized brass. Other equipment shall include a brass spillage pan, a 9 inch diameter brass base plate with leveling screws, a small spirit level, a spouted container which will hold approximately 550 grams of powder, and a weighing scale accurate to 0.1 gram. The brass spillage pan shall be of sufficient size to serve as a container for the apparatus and a device for plumbing the assembled apparatus. The cylindrical sections shall be machined to fit one above the other and the screen and distribution plate machined so they can be inserted between these sections. The plates shall contain holes that are leveled on both sides so that either side will allow powder to flow freely when inserted between the tubes.

4.5.6.2 Assembly. The apparatus shall be assembled as follows (see Figures 1 and 2): The circular base plate shall be placed in the spillage pan. This plate contains a circular area of the same outside diameter as the bottom of the 500 ml cylinder. The 500 ml cylinder shall be seated in the circular area. The small holed loading screen shall be placed on top of tube number 3, the sectional tube number 2 shall be placed on top of number 3, and this unit set on top of the 500 ml cylinder. The brass distribution plate shall be placed on tube section number 2 and then tube number 1 shall be added. After the numbered tubes and plates are mounted correctly, the funnel-shaped hopper shall be placed into tube number 1. The spillage pan shall be grounded to a cold water pipe with a grounding cable, the spirit level placed on the base plate, and the leveling screws adjusted until the plate is level and solid.

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4.5.6.3 Conditioning. The sample of casting powder shall be conditioned to a constant temperature of 80°F. The casting powder does not have to be conditioned for humidity provided the moisture content of the sample is between 0.1 and 3.0 percent and the test is made in a room having a relative humidity between 30 and 80 percent.

4.5.6.4 Procedure. The empty 500 ml cylinder shall be tared, the weight recorded, and the apparatus assembled as specified in 4.5.6.2. The spouted container shall be tared and approximately 545 grams of powder weighed out. The powder shall be dumped into the hopper in one motion. When all the powder has run through the column, all parts of the apparatus above the 500 ml cylinder shall be removed. This shall be accomplished by holding the bottom cylinder firmly and lifting off the other sections as a unit. Excess powder shall be struck off by using the brass bar with the 1/4 inch edge held flush with the top of the cylinder and with a slow constant movement across the cylinder top toward the operator. The bar shall not be slanted nor excess powder struck off again. The cylinder shall be removed from the base plate and any powder, that may be clinging to the sides of the container, brushed off. The screen loading density shall be calculated to determine compliance with 3.6 by subtracting the tare weight of the cylinder from the gross weight of the powder and cylinder and multiplying the difference by 0.002.

4.5.7 Grain dimensions, Major Defect. The grain dimensions shall be determined in accordance with Method 504.4 of MIL-STD-286.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Packing. The base grain shall be packed in accordance with DOT regulations and the best commercial practice.

5.2 Marking. The containers shall be marked in accordance with DOT regulations and MIL-STD-129.

6. NOTES

6.1 Intended use. The AHH casting powder is intended for use in the 155MM Rocket Assisted Projectile (M549) Propellant Grain.

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6.2 ORDERING DATA. See 6.1.2 of MIL-A-48078 (AR).

6.3 Submission of AMC Form 1047. See 6.1.2 of MIL-A-48078 (AR).

6.4 PREVIOUE ISSUES. Previous issues of MIL-P-60432, having been issued as classified documents strictly for use of preparing activity, and, therefore, are not available from Department of Defense Single Stock Point.

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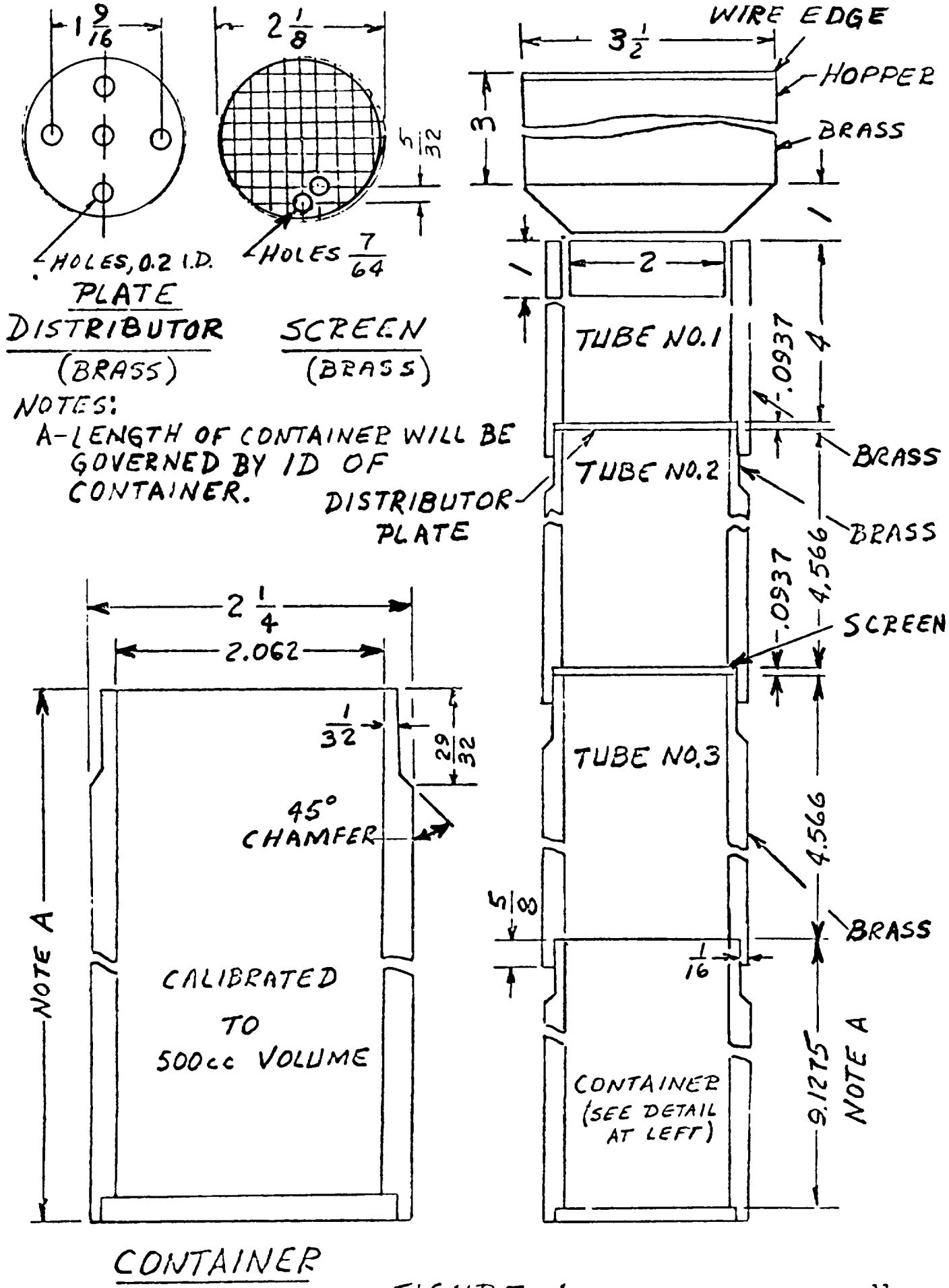


FIGURE 1

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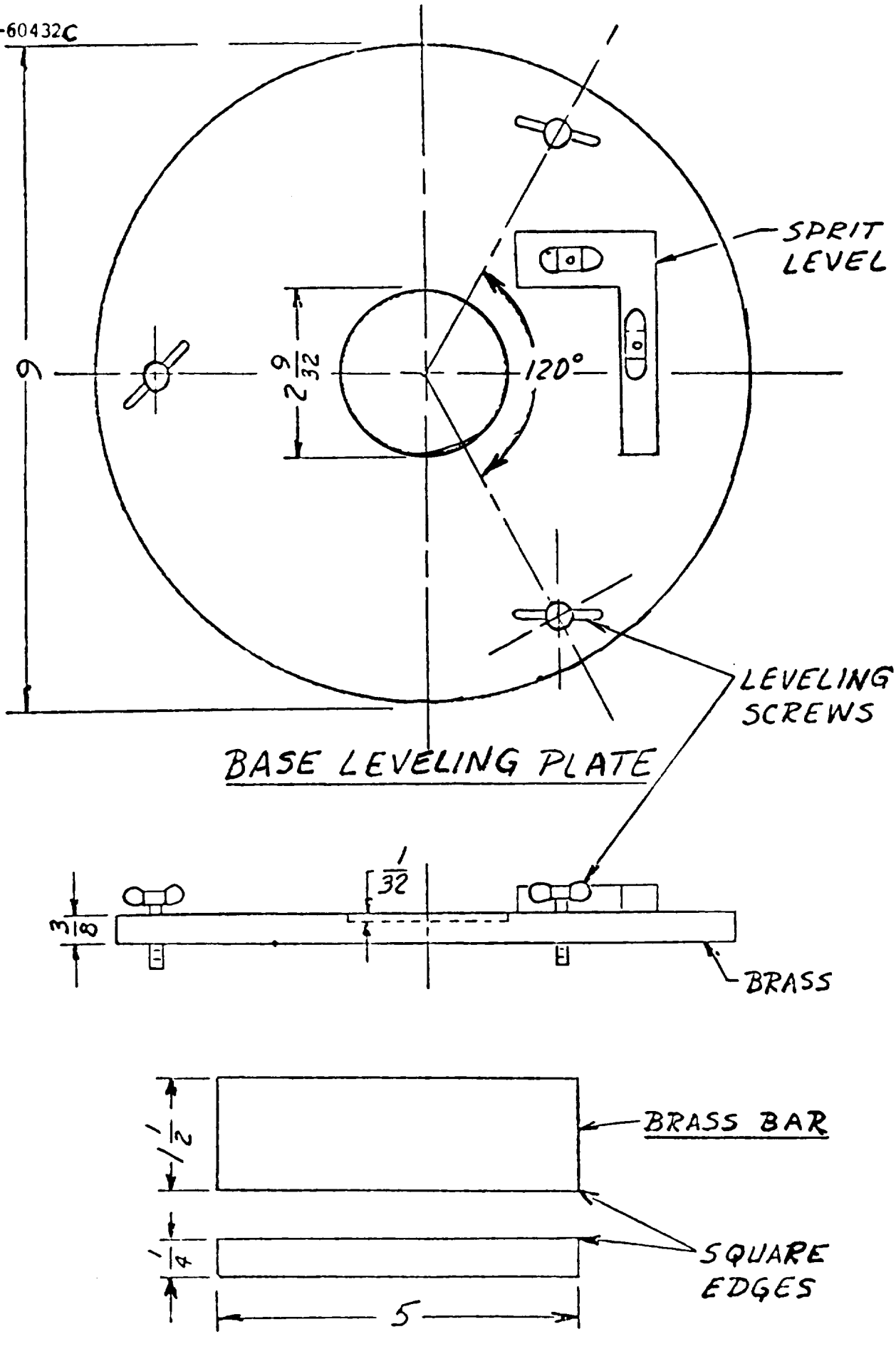


FIGURE 2

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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